

Total No. of Questions : 8]

SEAT No. :

PB2504

[6263]-390

[Total No. of Pages :2

**B.E. (Artificial Intelligence & Data Science)
COMPUTATIONAL INTELLIGENCE
(2019 Pattern) (Semester-VIII) (417530)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicates full marks.

- Q1)** a) What are the key performance measures used to evaluate the effectiveness of evolutionary algorithms? [6]
- b) What is swarm intelligence, and how does it differ from traditional optimization techniques? [6]
- c) Define Genetic programming and its application in symbolic regression? [6]

OR

- Q2)** a) Discuss the importance of selection in evolutionary algorithms and its impact on the convergence of the algorithm? [6]
- b) Explain the concept of population in evolutionary computing and its significance in the optimization processes? [6]
- c) Compare and contrast genetic algorithms, evolution strategies, evolutionary programming. [6]

- Q3)** a) Explain the concept of an individual in the context of genetic algorithms and its role in the optimization process? [6]
- b) Explain the concept of Messy Genetic Algorithm? [6]
- c) Write the principles of the Holland Classifier system and its application? [5]

OR

- Q4)** a) Describe the process of selection in genetic algorithms, including the different selection strategies used? [6]
- b) What condition determines when a genetic algorithm stops iterating and returns the best solution found? [6]
- c) Explain the concept of initialization in genetic algorithms and its role in creating the initial population? [5]

P.T.O.

Q5) a) Compare and contrast Word2Vec and GloVe in terms of how they generate word embeddings? [9]

b) Explain the architecture of a Seq2Seq model and its role in neural machine translation? [9]

OR

Q6) a) How does BERT (Bidirectional Encoder Representations from Transformers) work, and what are its advantages over traditional language models? [9]

b) Describe the TF-IDF (Term Frequency-Inverse Document Frequency) weighting scheme and its significance in text representation? [9]

Q7) a) Explain the concept of Artificial Immune Models and their significance in computational intelligence? [9]

b) Describe the principles of Network Theory Model and its application in understanding immune system behaviour? [8]

OR

Q8) a) Discuss the role of antigen-presenting cells in immune activation and its representation in the Dendritic Cell Model? [9]

b) Discuss the limitations and challenges of applying Artificial Immune System models in real-world applications? [8]

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